

USANDO

$$M^{\mu\nu} = i(x^\mu \partial^\nu - x^\nu \partial^\mu)$$

VALUTIAMO

$$[M^{\mu\nu}, M^{\rho\sigma}]$$

PER NON

PERDERE

PEZZI

SERVIAMOCI

DI

UNA FUNZIONE

$$f = f(x^\mu);$$

$$\begin{aligned} M^{\mu\nu} M^{\rho\sigma} f &= -i(x^\mu \partial^\nu - x^\nu \partial^\mu)(x^\rho \partial^\sigma - x^\sigma \partial^\rho) f \\ &= -[x^\mu \partial^\nu (x^\rho \partial^\sigma) - x^\mu \partial^\nu (x^\sigma \partial^\rho) - x^\nu \partial^\mu (x^\rho \partial^\sigma) + x^\nu \partial^\mu (x^\sigma \partial^\rho)] f(x) \\ &= -[x^\mu (\partial^\nu x^\rho \partial^\sigma + x^\rho \partial^\nu \partial^\sigma) - x^\mu (\partial^\nu x^\sigma \partial^\rho + x^\sigma \partial^\nu \partial^\rho) - \\ &\quad - x^\nu (\partial^\mu x^\rho \partial^\sigma + x^\rho \partial^\mu \partial^\sigma) + x^\nu (\partial^\mu x^\sigma \partial^\rho + x^\sigma \partial^\mu \partial^\rho)] \end{aligned}$$

$$\begin{aligned} \partial^\nu &= g^{\nu\alpha} \partial_\alpha \\ \partial^\nu x^\rho &= g^{\nu\alpha} \partial_\alpha x^\rho \\ &= g^{\nu\alpha} \delta_\alpha^\rho \\ &= g^{\nu\rho} \end{aligned}$$

IN QUESTO MODO:

$$[M^{\mu\nu}, M^{\rho\sigma}] f = [M^{\mu\nu} M^{\rho\sigma} - M^{\rho\sigma} M^{\mu\nu}] f =$$

$$\begin{aligned} &= -[x^\mu \underline{g^{\nu\rho} \partial^\sigma} + \underline{x^\mu x^\rho \partial^\nu \partial^\sigma} - \underline{x^\mu g^{\nu\sigma} \partial^\rho} - \underline{x^\mu x^\sigma \partial^\nu \partial^\rho} - \underline{x^\nu g^{\mu\rho} \partial^\sigma} - \underline{x^\nu x^\rho \partial^\mu \partial^\sigma} + \\ &\quad + \underline{x^\nu g^{\mu\sigma} \partial^\rho} + \underline{x^\nu x^\sigma \partial^\mu \partial^\rho}] f + [\underline{x^\rho g^{\nu\mu} \partial^\sigma} + \underline{x^\rho x^\mu \partial^\nu \partial^\sigma} - \underline{x^\rho g^{\nu\sigma} \partial^\mu} - \underline{x^\rho x^\sigma \partial^\nu \partial^\mu} - \\ &\quad - \underline{x^\sigma g^{\mu\rho} \partial^\nu} - \underline{x^\sigma x^\rho \partial^\mu \partial^\nu} + \underline{x^\sigma g^{\nu\mu} \partial^\rho} + \underline{x^\sigma x^\nu \partial^\mu \partial^\rho}] f \end{aligned}$$

VISTA LA
SIMMETRIA
DI $g^{\mu\nu}$ TERMINI
RACCOLGERE:

SOTTOLINEATI A ZIG-ZAG SI CANCELLANO, MENTRE GLI ALTRI LI POSSIAMO

$$[M^{\mu\nu}, M^{\rho\sigma}] = -[g^{\nu\rho} (x^\mu \partial^\sigma - x^\sigma \partial^\mu) + g^{\nu\sigma} (x^\rho \partial^\mu - x^\mu \partial^\rho) + \dots]$$

(1)

$$\dots + g^{\mu\rho}(x^\sigma\partial^\nu - x^\nu\partial^\sigma) + g^{\mu\sigma}(x^\nu\partial^\rho - x^\rho\partial^\nu)$$

POSSIAMO RICONOSCERE : $-i H^{\mu\nu} = (x^\mu\partial^\nu - x^\nu\partial^\mu)$

$$[H^{\mu\nu}, H^{\rho\sigma}] = -i [g^{\nu\rho} H^{\mu\sigma} + g^{\nu\sigma} H^{\rho\mu} + g^{\mu\rho} H^{\sigma\nu} + g^{\mu\sigma} H^{\nu\rho}]$$

POSSIAMO ANCHE SISTEMARE I SEGNI USANDO LA SIMMETRIA DI $g^{\mu\nu}$ E L'ANTISIMMETRIA DI $H^{\mu\nu}$. LA STRUTTURA DEGLI INDICI DEI TERMI È :

$$[12, 34] = -i [(14, 23) + (23, 14) - (13, 24) - (24, 13)]$$

$\swarrow$ INDICI g $\searrow$ INDICI H

SISTEMANDO LA NOSTRA SCRITTURA PER OTTENERE :

$$[H^{\mu\nu}, H^{\rho\sigma}] = -i [(\mu\sigma, \nu\rho) + (\nu\rho, \mu\sigma) - (\mu\rho, \nu\sigma) - (\nu\sigma, \mu\rho)]$$

NOI ABBIAMO

$$= -i [g^{\mu\sigma} H^{\nu\rho} + g^{\nu\rho} H^{\mu\sigma} - g^{\mu\rho} H^{\nu\sigma} - g^{\nu\sigma} H^{\mu\rho}]$$

AVENDO

$$N_i = \frac{1}{2}(\mathcal{J}_i + i K_i)$$

$$N_i^\dagger = \frac{1}{2}(\mathcal{J}_i - i K_i)$$

E

FACILE

VERIFICARE

VALGONO!

$$[\mathcal{J}_i, \mathcal{J}_j] = i \varepsilon_{ijk} \mathcal{J}_k$$

$$[K_i, K_j] = -i \varepsilon_{ijk} \mathcal{J}_k$$

$$[\mathcal{J}_i, K_j] = i \varepsilon_{ijk} K_k$$

$$[N_i, N_j^\dagger] = \left[\frac{1}{2}(\mathcal{J}_i + i K_i), \frac{1}{2}(\mathcal{J}_j - i K_j) \right]$$

$$= \frac{1}{4} \left([\mathcal{J}_i, \mathcal{J}_j] - i [\mathcal{J}_i, K_j] + i [K_i, \mathcal{J}_j] + [K_i, K_j] \right)$$

$$= \frac{1}{4} \left(i \varepsilon_{ijk} \mathcal{J}_k + \varepsilon_{ijk} K_k + i \underbrace{(-i \varepsilon_{ijk} K_k)}_{-[\mathcal{J}_j, K_i] = i \varepsilon_{jki} \mathcal{J}_k} - i \varepsilon_{ijk} \mathcal{J}_k \right)$$

$$= \frac{1}{4} \left(i \varepsilon_{ijk} \mathcal{J}_k + \varepsilon_{ijk} K_k - \varepsilon_{ijk} K_k - i \varepsilon_{ijk} \mathcal{J}_k \right)$$

$$= 0$$

$$\begin{aligned} [A, B] &= AB - BA \\ [B, A] &= BA - AB \\ &= -[A, B] \end{aligned}$$

$$[N_i, N_j] = \frac{1}{4} [\mathcal{J}_i + i K_i, \mathcal{J}_j + i K_j] = \frac{1}{4} \left([\mathcal{J}_i, \mathcal{J}_j] + i [\mathcal{J}_i, K_j] + i [K_i, \mathcal{J}_j] - [K_i, K_j] \right)$$

$$= \frac{1}{4} \left(i \varepsilon_{ijk} \mathcal{J}_k + i (\varepsilon_{ijk} K_k) + i (-\varepsilon_{jik} K_k) + \varepsilon_{ijk} \mathcal{J}_k \right)$$

$$= \frac{1}{2} i \varepsilon_{ijk} \mathcal{J}_k - \frac{1}{2} \varepsilon_{ijk} K_k$$

$$= i \varepsilon_{ijk} \left(\frac{1}{2} (\mathcal{J}_k + i K_k) \right) = i \varepsilon_{ijk} N_k$$

$$[N_i^+, N_j^+] \cdot \frac{1}{4} [J_i - iK_i, J_j - iK_j]$$

$$= \frac{1}{4} \left([J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] - [K_i, K_j] \right)$$

$$\underbrace{- [J_j, K_i]}_{= -i\varepsilon_{jik} K_k = i\varepsilon_{ijk} K_k}$$

$$= \frac{1}{4} \left(i\varepsilon_{ijk} J_k - i(i\varepsilon_{ijk} K_k) - i(i\varepsilon_{ijk} K_k) + i\varepsilon_{ijk} K_k \right)$$

$$= \frac{1}{2} i\varepsilon_{ijk} J_k + \frac{1}{2} i\varepsilon_{ijk} K_k$$

$$= i\varepsilon_{ijk} \frac{1}{2} (J_k - iK_k)$$

$$= i\varepsilon_{ijk} N_k^+$$

CONOSCO

$$(M^{\mu\nu})^\mu{}_\nu = -\lambda (g^{\mu\mu} \delta^\nu{}_\nu - g^{\mu\nu} \delta^\nu{}_\mu)$$

POSSIAMO

DETERMINARE

LE

6

MATRICI

$M^{\mu\nu}$.

$$(M^{01})^\mu{}_\nu = -\lambda (g^{\mu 0} \delta^1{}_\nu - g^{\mu 1} \delta^0{}_\nu) = -\lambda \left[\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} (0 \ 1 \ 0 \ 0) - \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} (1 \ 0 \ 0 \ 0) \right] = -\lambda \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(M^{02})^\mu{}_\nu = -\lambda (g^{\mu 0} \delta^2{}_\nu - g^{\mu 2} \delta^0{}_\nu) = -\lambda \left[\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} (0 \ 0 \ 1 \ 0) - \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix} (1 \ 0 \ 0 \ 0) \right] = -\lambda \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(M^{03})^\mu{}_\nu = -\lambda (g^{\mu 0} \delta^3{}_\nu - g^{\mu 3} \delta^0{}_\nu) = -\lambda \left[\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} (0 \ 0 \ 0 \ 1) - \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} (1 \ 0 \ 0 \ 0) \right] = -\lambda \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$(M^{12})^\mu{}_\nu = -\lambda (g^{\mu 1} \delta^2{}_\nu - g^{\mu 2} \delta^1{}_\nu) = -\lambda \left[\begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} (0 \ 0 \ 1 \ 0) - \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix} (0 \ 1 \ 0 \ 0) \right] = -\lambda \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(M^{23})^\mu{}_\nu = -\lambda (g^{\mu 2} \delta^3{}_\nu - g^{\mu 3} \delta^2{}_\nu) = -\lambda \left[\begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix} (0 \ 0 \ 0 \ 1) - \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} (0 \ 0 \ 1 \ 0) \right] = -\lambda \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & +1 & 0 \end{pmatrix}$$

$$(M^{13})^\mu{}_\nu = -\lambda (g^{\mu 1} \delta^3{}_\nu - g^{\mu 3} \delta^1{}_\nu) = -\lambda \left[\begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} (0 \ 0 \ 0 \ 1) - \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} (0 \ 1 \ 0 \ 0) \right] = -\lambda \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

(5)

POSSIAMO VERIFICARE CHE

$$J_i = \frac{1}{2} \varepsilon_{ijk} M^{jk}$$
 (UGUALE A $J_i = \frac{1}{2} \varepsilon_{ijk} M^{jk}$ SPAZIALI ESSE POICHÈ ABBASSANDO 2) "(-)(-) = +"

$$J_i = \frac{1}{2} \varepsilon_{ijk} M^{jk}$$

$$K^i = M^{0i}$$

E IMMEDIATO

$$K^1 = K_x = M^{01} = -\dot{\lambda} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$K^2 = K_y = M^{02} = -\dot{\lambda} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$K^3 = K_z = M^{03} = -\dot{\lambda} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

VEDI
ILIPOPOULOS
PAG. 94

$$J^1 = J^x = \frac{1}{2} \varepsilon^{ijk} M^{jk} = \frac{1}{2} \left(\varepsilon^{123} M^{23} + \underbrace{\varepsilon^{132} M^{32}}_{(-\varepsilon^{123})(-M^{23})} \right) = \underbrace{\varepsilon^{123}}_{+1} M^{23} = -\dot{\lambda} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & +1 & 0 \end{pmatrix}$$

$$J^2 = J^y = \frac{1}{2} \varepsilon^{2jk} M^{jk} = \frac{1}{2} \left(\underbrace{\varepsilon^{213} M^{13}}_{-\varepsilon^{123}} + \underbrace{\varepsilon^{231} M^{31}}_{\varepsilon^{123}(-M^{13})} \right) = -(M^{13}) = -\dot{\lambda} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & +1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

$$J^3 = J^z = \frac{1}{2} \varepsilon^{3jk} M^{jk} = \frac{1}{2} \left(\underbrace{\varepsilon^{312} M^{12}}_{\varepsilon^{123} M^{12}} + \underbrace{\varepsilon^{321} M^{21}}_{(-\varepsilon^{123})(-M^{12})} \right) = M^{12} = -\dot{\lambda} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(6)

VEDIAMO I MOMENTI ANGOLARI J_i E I BOOST K_i COSA DIVENTANO PER TRASFORMAZIONI DI PARITÀ.

VEDIAMO AD ESEMPIO J_1 :

$$\Lambda_P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

VEDI § 3.7.2
DI SCHWICHTENBERG

$$J_1' = \Lambda_P J_1 \Lambda_P^T = \Lambda_P J_1 \Lambda_P^T$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & -\lambda & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}^T$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & -\lambda & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & -\lambda & 0 \end{pmatrix} = J_1$$

ANALOGHI CONTI PER J_2 E J_3 . PER IL BOOST K_1 :

$$K_1' = \Lambda_P K_1 \Lambda_P^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -\lambda & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}^T = \begin{pmatrix} 0 & \lambda & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \lambda & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = -K_1$$

CONTI ANALOGHI PER K_2 , K_3 .

CON CONTI ANALOGHI SI VEDE PER L'INVERSIONE TEMPORALE:

$$J_i \xrightarrow{T} J'_i = \Lambda_T J_i \Lambda_T^T = J_i$$

$$K_i \xrightarrow{T} K'_i = \Lambda_T K_i \Lambda_T^T = -K_i$$

QUINDI COMPLESSIVAMENTE SCRIVIAMO

$$J_i \xrightarrow{P} J_i$$

$$K_i \xrightarrow{P} -K_i$$

$$J_i \xrightarrow{T} J_i$$

$$K_i \xrightarrow{T} -K_i$$

$$\Lambda_{LR} = e^{+\frac{i}{2}\sigma^i(\omega_i \mp \eta_i)}$$

$$\Lambda_{L,R}^* = e^{-\frac{i}{2}\sigma^i(\omega_i \pm \eta_i)}$$

$$\Lambda_{L,R}^T = e^{+\frac{i}{2}\sigma^{iT}(\omega_i \mp \eta_i)}$$

$$\Lambda_{L,R}^{-1} = e^{-\frac{i}{2}\sigma^i(\omega_i \mp \eta_i)}$$

$$\Lambda_{L,R}^+ = e^{-\frac{i}{2}\sigma^{i+}(\omega_i \pm \eta_i)}$$

$$\begin{cases} (\sigma^2)^* = -\sigma^2 & ; (\sigma^2)^T = -\sigma^2 \\ (\sigma^2)^T = \sigma^2 & ; (\sigma^i)^+ = \sigma^i \end{cases}$$

$$\sigma^2 \sigma^{-1} \sigma^2 = -\sigma^{i*} \Rightarrow -\sigma^i = \sigma^2 \sigma^{i*} \sigma^2$$

$$\textcircled{1} \Lambda_{L,R}^* \xrightarrow{+} \Lambda_{L,R}^+ = e^{-\frac{i}{2}\sigma^i(\omega_i \pm \eta_i)}$$

$$\Rightarrow \boxed{\Lambda_{L,R}^+ = \Lambda_{R,L}^{-1}}$$

$$\textcircled{2} \Lambda_{L,R}^* = \sigma^2 \Lambda_{R,L} \sigma^2$$

$$\textcircled{2} \sigma^2 \Lambda_{L,R}^* \sigma^2 = e^{+\frac{i}{2}\sigma^i(\omega_i \pm \eta_i)} = \Lambda_{R,L}$$

$$\textcircled{4} \textcircled{3} \xrightarrow{+ = * + T} (\Lambda_{L,R}^*)^+ = (\sigma^2 \Lambda_{R,L} \sigma^2)^+ \Rightarrow \Lambda_{L,R}^T = \sigma^2 \Lambda_{R,L}^+ \sigma^2 \xRightarrow{\textcircled{1}} \boxed{\Lambda_{L,R}^T = \sigma^2 \Lambda_{L,R}^{-1} \sigma^2}$$

$$\textcircled{5} \boxed{\Lambda_L^T \sigma^2 \Lambda_L = \Lambda_L^T \sigma^2 \Lambda_L \sigma^2 \sigma^2 = \Lambda_L^T \Lambda_R^* \sigma^2 = \sigma^2 \Lambda_L^{-1} \sigma^2 \Lambda_R^* \sigma^2 = \sigma^2 \Lambda_L^{-1} \Lambda_L = \sigma^2}$$

9

CONTO
GUARDO

GENERICO PER $\phi(x^\mu)$, $\mathcal{L}(\phi, \partial_\mu \phi)$

$$\Delta S = \int d^4x \Delta \mathcal{L} + \int \delta(d^4x) \mathcal{L}$$

$$\text{so } \delta(d^4x) = \det \delta x^\mu d^4x$$

PER UNA TRASFORMAZIONE
DI COORDINATE (VEG
NOTE)

CALCOLO

$$\begin{aligned} \Delta \mathcal{L} &= \delta \mathcal{L} + \delta x^\mu \partial_\mu \mathcal{L} \\ &= \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta (\partial_\mu \phi) + \delta x^\mu \partial_\mu \mathcal{L} \\ &= \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \partial_\mu (\delta \phi) + \delta x^\mu \partial_\mu \mathcal{L} \\ &= \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta \phi \right) - \partial_\mu \frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta \phi + \delta x^\mu \partial_\mu \mathcal{L} \end{aligned}$$

μ NETTO ON-SHELL $\Rightarrow$ EQ MOTO = 0

$$\Rightarrow \Delta \mathcal{L} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta \phi \right) + \delta x^\mu \partial_\mu \mathcal{L}$$

così

$$\Delta S = \int d^4x \left[\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta \phi \right) + \underbrace{\delta x^\mu \partial_\mu \mathcal{L} + \partial_\mu \delta x^\mu \mathcal{L}}_{\partial_\mu (\delta x^\mu \mathcal{L})} \right]$$

$$\Rightarrow \Delta S = \int d^4x \underbrace{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta \phi + \delta x^\mu \mathcal{L} \right)}_{J^\mu} = 0$$

QUINDI ABBIAMO

$$J^\mu = \frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta \phi + \delta x^\mu \mathcal{L}$$

SE DIFFERENZIO IL CASO DI TRASFORMAZIONE DI COORDINATE:

$$x^\mu \rightarrow x^\mu + \delta x^\mu \Rightarrow \Delta \phi = 0 \Rightarrow \underline{\delta \phi = -\delta x^\mu \partial_\mu \phi}$$

SOSTITUISCO

$$J^\mu = -\frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \delta x^\nu \partial_\nu \phi + \delta x^\mu \mathcal{L} = \left(-\frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \partial_\nu \phi + g^\mu{}_\nu \mathcal{L} \right) \delta x^\nu = \boxed{-T^\mu{}_\nu \delta x^\nu}$$

TENSORE
ENERGIA-IMPULSO

VEDIAMO ANCHE IL CASO GENERICO:

$$x^\mu \rightarrow x^\mu + \delta x^\mu, \quad \phi \rightarrow \phi + \alpha \Rightarrow \underline{\Delta \phi = \delta \phi + \delta x^\mu \partial_\mu \phi}$$

SOSTITUENDO ~~$\delta \phi$~~ $\delta \phi = \Delta \phi - \delta x^\mu \partial_\mu \phi$

$$J^\mu = -T^\mu{}_\nu \delta x^\nu + \frac{\partial \mathcal{L}}{\partial [\partial_\mu \phi]} \Delta \phi$$

$$\delta S = S(x') - S(x) = \int d^4x (L'(x) - L(x)) = \int d^4x \delta L$$

$$d^4x' = (1 + \partial_\mu \delta x^\mu) d^4x$$

$$\Delta S = S'(x') - S(x) = \int d^4x' L'(x') - \int d^4x L(x) \quad L'(x) = \Delta L + L(x)$$

$$= \int (1 + \partial_\mu \delta x^\mu) (\Delta L + L(x)) d^4x - \int d^4x L(x)$$

$$\approx \int d^4x (\Delta L + \partial_\mu \delta x^\mu L) = \int d^4x (\delta L + \delta x^\mu \underbrace{\partial_\mu L}_{\partial_\mu (\delta x^\mu L)})$$

$$\Delta L = \delta L + \delta x^\mu \partial_\mu L$$

$$\partial_\mu (\delta x^\mu L)$$

$$\Delta S = \delta S + \int d^4x \partial_\mu (L \delta x^\mu)$$

(12)